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1. Let $x, y \in \mathbb{R}$. Which of the following statements is true based on the properties of real numbers?
- A) If $x < y$, then $x^2 < y^2$ for all real x and y .
B) If $x^2 < y^2$, then $x < y$ for all real x and y .
C) If $x < y$, then $x^2 \leq y^2$ for all real x and y .
D) If $x < y$ and $x, y \geq 0$, then $x^2 < y^2$.
2. The intersection of all intervals $(-1/n, 1/n)$, $n = 1, 2, \dots$ is the set
- A) null set B) $\{0\}$ C) $(-\epsilon, +\epsilon)$ where ϵ is very small but positive D) $\{1/n\}$
3. If $A + B + C = 180^\circ$, then $\cos(A + B)$ is equal to:
- A) $\cos A - \cos B$ B) $\cos A \cdot \cos B$ C) $-\cos C$ D) $\sin A + \sin B$
4. A line passes through the point $(2, 3)$ and is perpendicular to the line $3x - 4y + 5 = 0$. What is the equation of the line?
- A) $3x + 4y = 17$ B) $4x - 3y = 17$ C) $4x + 3y = 17$ D) $4x - 3y = 5$
5. The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. The rate at which the area increases when the side is 10 cm is
- A) $10 \text{ cm}^2/\text{sec}$ B) $10\sqrt{3} \text{ cm}^2/\text{sec}$ C) $\sqrt{3} \text{ cm}^2/\text{sec}$ D) $\frac{10}{\sqrt{3}} \text{ cm}^2/\text{sec}$
6. Value of the integral $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} d\theta$ is equal to
- A) $2(\cos x + \theta \sin \theta) + C$ B) $2(\cos x - x \sin \theta) + C$
C) $2(\sin x - x \cos \theta) + C$ D) $2(\sin \theta + \theta \cos x) + C$
7. In how many ways can 5 people be seated in a row if two specific people must not sit next to each other?
- A) 48 B) 72 C) 96 D) 108

8. If the function $f(x)$ is defined by the rule $f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$ then which among the following is true?
- A) $f(x)$ is of bounded variation on the set of rational points in $[0, 1]$
 B) $f(x)$ is of bounded variation on the set of irrational points in $[0, 1]$
 C) $f(x)$ is of bounded variation on any interval
 D) $f(x)$ is not of bounded variation on any interval
9. If a function f is monotonic on $[a, b]$, then the set of discontinuities of f is
- A) empty B) finite C) countable D) $[a, b]$
10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Which of the following statements is true?
- A) f is differentiable everywhere on $\mathbb{R}$
 B) f is integrable on $\mathbb{R}$
 C) f attains a maximum and minimum on any closed interval
 D) f is bounded on $\mathbb{R}$
11. The sequence $\{S_n\}$ defined by $S_{n+1} = \sqrt{7 + S_n}$ and $S_1 = \sqrt{7}$ converges to
- A) positive root of $x^2 - x - 7 = 0$
 B) negative root of $x^2 - x - 7 = 0$
 C) positive root of $x^2 + x - 7 = 0$
 D) negative root of $x^2 + x - 7 = 0$
12. For the complex number i , $\frac{3i^{30} - i^{19}}{2i - 1}$ is equal to
- A) i B) $-i$ C) $1 + i$ D) $1 - i$
13. $\int_C \frac{\cos z}{z} dz$ where C is the unit circle is
- A) $2\pi i$ B) πi C) $\frac{\pi}{2}i$ D) none of these
14. Residue of the function $\cot z$ at the singular points is
- A) 0 B) i C) $-i$ D) 1

15. The value of $\int_C \frac{z^2 - 4}{z(z^2 + 9)} dz$ where C is the circle $|z| = 1$ is
 A) $\frac{-2\pi i}{9}$ B) $\frac{-4\pi i}{9}$ C) $\frac{-8\pi i}{9}$ D) none of these
16. Let $G = \langle a \rangle$ be a cyclic group of order 60. How many generators does G have?
 A) 16 B) 32 C) 60 D) 1
17. The units of $\mathbb{Z}_4$, the set of integers modulo four, are
 A) 1 B) 1,3 C) 1,2,3 D) none of these
18. The characteristic of the ring $\mathbb{Z} \times \mathbb{Z}$ is
 A) 0 B) 1 C) 2 D) none of these
19. Which among the following numbers is **not** a solution of the equation $x^3 - 2x^2 - 3x = 0$ in $\mathbb{Z}_{12}$
 A) 0 B) 2 C) 3 D) 11
20. The remainder of 3^{47} when divided by 23 is
 A) 3 B) 4 C) 7 D) 12
21. A Sylow 3-subgroup of a group of order 12 has order
 A) 1 B) 2 C) 3 D) 4
22. The matrix $A = \begin{pmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{pmatrix}$ is singular if
 A) $x = 0$ B) $x = 2$ C) $x = 4$ D) $x = 8$
23. If A and B are symmetric matrices of the same order, then which of the following is true?
 A) AB is always symmetric B) AB is never symmetric
 C) AB is skew symmetric D) AB is symmetric if and only if $AB = BA$
24. The value of the determinant $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$ is equal to
 A) $a + b + c$ B) abc C) zero D) none of these

25. If V is a vector space of finite dimension and W is a subspace of V , then $\dim V/W$ is equal to
 A) $\dim V + \dim W$ B) $\dim V - \dim W$ C) $\dim V \times \dim W$ D) none of these
26. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a transformation defined by $T(x, y) = (x + y, x - y)$. Then T is:
 A) not a linear transformation
 B) a linear transformation but not one-to-one
 C) a linear transformation but not onto
 D) a linear transformation and is both one-to-one and onto
27. Which of the following sets is a vector space over $\mathbb{R}$ under usual addition and scalar multiplication?
 A) The set of all 2×2 matrices with determinant equal to 1
 B) The set of all non-zero real numbers
 C) The set of all polynomials of degree exactly 3
 D) The set of all polynomials of degree less than or equal to 3
28. If T is a linear operator on a finite dimensional vector space V such that 0 is a characteristic value of T , then
 A) T is invertible B) T is not invertible
 C) $Tv = 0$ for all $v \in V$ D) none of these is correct
29. The value of $\phi(999)$, where ϕ is the Euler's totient function is
 A) 729 B) 666 C) 648 D) 576
30. By eliminating the arbitrary constants a and b from the equation $y = ae^x + bxe^x$ we get the differential equation
 A) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$ B) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$
 C) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x$ D) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$
31. Particular integral of the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = x$ is
 A) $\frac{6x - 1}{36}$ B) $-\frac{6x + 1}{36}$ C) $-\frac{6x - 1}{36}$ D) none of these

32. The general solution of the equation $x(y - z)\frac{\partial z}{\partial x} + y(z - x)\frac{\partial z}{\partial y} = z(x - y)$ will be
- A) $F(x + y + z, xyz) = 0$ B) $F(xy + yz, xyz) = 0$
 C) $F(x^2 - y^2, z) = 0$ D) none of these
33. Orthogonal trajectories of the family of parabolas $y^2 = 4ax$ will be the
- A) family of lines $y = mx$, passing through the origin
 B) family of circles $x^2 + y^2 = a^2$
 C) family of curves $2x^2 + y^2 = c$
 D) family of rectangular hyperbolas $xy = c^2$
34. Let X be a topological space and $A \subseteq X$. Which of the following statements about the closure of A , denoted $\overline{A}$, is false?
- A) $\overline{A}$ is the smallest closed set containing A B) $\overline{A}$ contains all the limit points of A
 C) A is closed if and only if $A = \overline{A}$ D) If A is open, then $\overline{A}$ is open
35. Which of the following statements is true about compact subsets of $\mathbb{R}^n$?
- A) Every compact set in $\mathbb{R}^n$ is open and bounded
 B) Every compact set in $\mathbb{R}^n$ is closed but not necessarily bounded
 C) Every compact set in $\mathbb{R}^n$ is open but not necessarily bounded
 D) Every compact set in $\mathbb{R}^n$ is closed and bounded
36. In a general topological space, which of the following statements is always true about compact sets?
- A) Every compact set is closed and bounded
 B) Every compact set is open
 C) Every compact set is finite
 D) The continuous image of a compact set is compact

37. Which of the following is a defining property of a Hausdorff topological space?
- A) Every open cover has a finite subcover
 - B) Every sequence has a convergent subsequence
 - C) For any two distinct points, there exist disjoint open sets containing each point
 - D) Every compact set is open
38. Let $\vec{u} = (1, 0)$ and $\vec{v} = (0, 1)$ in the inner product space $\mathbb{R}^2$ with the standard inner product. What is the angle θ between $\vec{u}$ and $\vec{v}$?
- A) 0°
 - B) 30°
 - C) 90°
 - D) 45°
39. If α and β are orthogonal unit vectors in an inner product space, what is the distance between α and β ?
- A) 1
 - B) 2
 - C) $\sqrt{2}$
 - D) 0
40. The value of $\lim_{x \rightarrow 1} \left[\frac{1}{\log x} - \frac{1}{x-1} \right]$ is
- A) $\frac{1}{2}$
 - B) 0
 - C) 1
 - D) -1
41. The point on the parabola $y^2 = 4x$ closest to the circle $x^2 + y^2 - 24y + 128 = 0$ is
- A) $(4, -4)$
 - B) $(4, 4)$
 - C) $(0, 0)$
 - D) $(2, 2\sqrt{2})$
42. The solution of $(1 + y^2)dx = (\tan^{-1}y - 2y)dy$ satisfying $y(0) = 1$ is
- A) $8\tan^{-1}y = e^{2x}(1 + y^2)$
 - B) $8\tan^{-1}y = e^x(1 + y^2)$
 - C) $4\tan^{-1}y = e^x(1 + y^2)$
 - D) $8\tan^{-1}y = \pi e^x(1 + y^2)$
43. The differential equation representing all parabolas having their axis of symmetry coinciding with the x -axis is
- A) $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 0$
 - B) $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$
 - C) $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right) = 0$
 - D) $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right) = 0$

44. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$, then $\lim_{x \rightarrow 0} f'(\frac{1}{x})$ is
- A) 0 B) $\frac{1}{2}$ C) 1 D) -1
45. The minimal polynomial for the matrix $A = \begin{bmatrix} 2 & 7 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -2 & 4 \end{bmatrix}$
- A) $(\lambda - 2)(\lambda - 3)$ B) $(\lambda - 2)(\lambda - 3)^2$
 C) $(\lambda - 2)^2(\lambda - 3)$ D) $(\lambda - 2)^2(\lambda - 3)^2$
46. Consider the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x + y, x + 2y)$. The inverse S of T is given by
- A) $S(x, y) = (2x - y, -x + y)$ B) $S(x, y) = (2x + y, x + y)$
 C) $S(x, y) = (x + 2y, x + y)$ D) None of these
47. Let V be a real vector space and let $\{x_1, x_2, x_3\}$ be a basis of V . Then
- A) $\{x_1 + x_2, x_2, x_3\}$ is a basis of V .
 B) $\{x_1 + x_2, x_2 + x_3, x_1 - x_3\}$ is a basis of V .
 C) $\{x_1 - x_2, x_2 - x_3, x_1 - x_3\}$ is a basis of V .
 D) None of these.
48. The rank of the matrix $A = \begin{bmatrix} 1 & 2 & k \\ -1 & 1 & 2 \\ 3 & 1 & 2 \end{bmatrix}$ is 2, then the possible value of k is
- A) 1 B) 4 C) -4 D) -1

49. Consider the following system of simultaneous equations: $x + 2y + z = 8$, $2x + y + 3z = 13$, $3x + 4y - \lambda z = \mu$, then

- A) The system has unique solution if $\lambda = \frac{-11}{3}$ and $\mu = 22$.
 B) The system has no solution if $\lambda = \frac{-11}{3}$ and $\mu \neq 22$.
 C) The system has infinite number of solutions if $\lambda = \frac{-11}{3}$.
 D) The system has infinite number of solutions if $\mu = 22$.

50. If for a matrix A of order 2, $A^2 + I = 0$, where I is the identity matrix of order 2, then A equals

- A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ B) $\begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$ C) $\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ D) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

51. The vector (a, b, c) belongs to the subspace of the vector space R^3 over R spanned by the vectors $(1, 2, 0)$ and $(-1, 1, 2)$, then

- A) $2a - 2b + 3c = 0$ B) $a + 2b - c = 0$
 C) $a + 2b - 3c = 0$ D) $2a - 3b + 2c = 0$

52. Consider the following statements:

- (I) Every linear operator on a real vector space has at least one eigen value.
 (II) If A is a real symmetric matrix, all eigen values are real.

- A) Both (I) and (II) are true. B) Only (I) is true.
 C) Only (II) is true D) Neither (I) nor (II) is true.

53. The value of m for which $u(x, y) = 2x - x^2 + my^2$ is harmonic is

- A) 0 B) -1 C) 1 D) 2

54. Let $f(z) = \exp\left(z + \frac{1}{z}\right)$ in $C - \{0\}$. Residue of $f(z)$ at $z = 0$ is

- A) $\sum_{k=0}^{\infty} \frac{1}{k!(k+1)!}$ B) $\sum_{k=0}^{\infty} \frac{1}{[k(k+1)]!}$
 C) $\sum_{k=0}^{\infty} \frac{1}{(k^2+1)!}$ D) $\sum_{k=0}^{\infty} \frac{1}{k!(k-1)!}$

55. The analytic function $f(z)$ whose real part is $u = 2x(1 - y)$
 A) $z^2 + 2z$ B) $iz^2 + 2z$ C) $iz^2 - 2z$ D) $z^2 + 2iz$
56. Let $\{x_n\}$ be the sequence defined by $x_n = \left(1 - \frac{3}{n}\right)^n$. The sequence $\{x_n\}$
 A) converges to 1 B) converges to e^3 C) converges to $\frac{1}{e^3}$ D) diverges.
57. The series $\sum_{n=1}^{\infty} \left(\frac{1}{2} + \frac{1}{3^n}\right)$
 A) converges to 1 B) converges to $\frac{3}{2}$ C) converges to $\frac{5}{6}$ D) diverges
58. Let $f : [0, 2] \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$
 and $\alpha : [0, 2] \rightarrow \mathbb{R}$ be defined by $\alpha(x) = x^3$. The $\int f d\alpha$ is
 A) 0 B) 1 C) 2 D) does not exist
59. Number of elements of order 40 in a cyclic group of order 200 is
 A) 16 B) 20 C) 40 D) 50
60. Number of onto group homomorphisms from the group $\mathbb{Z}_{20}$ to $\mathbb{Z}_{10}$ is
 A) 1 B) 4 C) 5 D) 10
61. Up to isomorphism, number of abelian groups of order 2025 is
 A) 10 B) 15 C) 20 D) 25
62. Let R be the polynomial ring $\mathbb{R}[x]$. Consider the ideals $I = \langle x^2 + 1 \rangle$ and $J = \langle x^4 + 1 \rangle$.
 Which of the following is true
 A) I is maximal ideal but not prime ideal. B) I is maximal ideal and prime ideal.
 C) J is a prime ideal. D) J is maximal but I is not maximal.
63. Which of the following is true about the ring $R = \mathbb{Z}[\sqrt{-5}]$
 A) R is a unique factorisation domain B) R is a euclidean domain
 C) R is a principal ideal domain D) $1 + 3\sqrt{-5}$ is irreducible in R
64. Degree of the splitting field of $x^5 - 1$ over $\mathbb{Q}$ is
 A) 1 B) 2 C) 4 D) 5

65. Which of the following is irreducible over $\mathbb{Z}_7[x]$
- A) $x^4 + x^2 + 2x + 3$ B) $x^3 + 3x + 2$
 C) $x^4 + 5x^2 + 2x + 2$ D) $x^4 + x^2 + 4x + 3$
66. The remainder obtained when 2^{2025} is divided by 13
- A) 5 B) 3 C) 2 D) 1
67. Number of incongruent solutions of the equation $6x \equiv 21 \pmod{30}$ is
- A) 3 B) 5 C) 6 D) 15
68. The Partial differential equation of the family of curves given by $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$, where a and b are arbitrary constants is
- A) $2z = px + qy$ B) $z = px - qy$ C) $z^2 = px - qy$ D) $py + qx = 0$
69. The characteristics of the Partial Differential equation $2u_{xx} - 4u_{xy} + 6u_{yy} + u_y = 0$
- A) $x + y = \text{constant}, 3x + y = \text{constant}$
 B) $x - y = \text{constant}, 3x - y = \text{constant}$
 C) $x + y = \text{constant}, 3x - y = \text{constant}$
 D) $x - y = \text{constant}, 3x + y = \text{constant}$
70. The equation $u_{xx} + 2yu_{xy} + xu_{yy} = 0$ is
- A) Hyperbolic everywhere.
 B) Hyperbolic in the region $y < x^2$.
 C) Hyperbolic in the region $y^2 < x$.
 D) Hyperbolic in the region $y^2 > x$.
71. Let $p(x) = x^3 - \beta x^2 - \alpha x + \alpha\beta$ for $\alpha \geq 0, \beta \geq 0$. Then, the roots of $p(x)$ are
- A) $\alpha, -\alpha, \beta$ B) $\alpha, \beta, -\beta$ C) $\alpha, \sqrt{\beta}, -\sqrt{\beta}$ D) $\sqrt{\alpha}, -\sqrt{\alpha}, \beta$
72. Equation of the line perpendicular to $2x + 3y = 5$ and passing through $(1, 1)$ is
- A) $3x + 2y = 1$ B) $3x + 2y = 5$
 C) $3x - 2y = 1$ D) $3x - 2y = 5$

73. The conic $x^2 + 3xy + 2y^2 + x - 2y + 5 = 0$ is a
- A) hyperbola B) ellipse C) parabola D) circle
74. For an examination, a student does not know the answers of 5 multiple choice questions and decides to choose one out of 4 choices randomly, for these 5 questions. Find the probability that he answers atleast one question correctly.
- A) $243/1024$ B) $781/1024$
 C) $992/1024$ D) $32/1024$
75. Which of the following is a closed set?
- A) $\{(x, y) : \sqrt{2} < x^2 + y^2 \leq \sqrt{3}\}$ in $\mathbb{R}^2$ with usual topology
 B) set of all polynomials on $[0, 1]$ with sup metric
 C) $\mathbb{Q}$ with discrete metric
 D) None of the above
76. Which of the following is **not** a metric on $\mathbb{R}$
- A) $d(x, y) = |x^2 - y^2| + |x - y|$ B) $d(x, y) = \max\{|x|, |y|\}$
 C) $d(x, y) = 2|x| + 2|y|$ D) $d(x, y) = |x - y||x + y|$
77. Which of the following is true?
- A) $\{\frac{1}{n}\}$ converges to 0 in $\mathbb{R}$ with discrete topology
 B) $\{n\}$ converges to $\sqrt{2}$ in $\mathbb{R}$ with co-finite topology
 C) $\{\frac{1}{n}\}$ does not converge to 0 in $\mathbb{R}$ with co-finite topology
 D) $(1, 0, 1, 0, 1, 0, \dots)$ converges to 0 in $\mathbb{R}$ with co-finite topology
78. Which among the following pairs of subsets of $\mathbb{R}^2$ are homeomorphic?
- A) $\{(x, y) : x^2 + y^2 = 1\}$ and $\{(x, y) : (x-1)^2 + y^2 = 1\} \cup \{(x, y) : (x+1)^2 + y^2 = 1\}$
 B) $\{(x, y) : xy = 1\}$ and $\{(x, y) : |x| = |y|\}$
 C) $\{(x, \sin x) : x \in \mathbb{R}\}$ and $\{(x, \cos x) : x \in \mathbb{R}\}$
 D) $\{(x, y) : x^2 + y^2 = 1\}$ and $\{(x, y) : 2x^2 - 3y^2 = 1\}$

79. Let $n \in \mathbb{N}$ be fixed and $T : \mathbb{C}^n \rightarrow l^2$ be a linear map. Then,

- A) $\|T\| \leq n$ B) T is injective
C) T is bounded below D) T is continuous

80. Let $\{u_n : n \in \mathbb{N}\}$ be an orthonormal basis of l^2 . Which of the following series is convergent in l^2 ?

- A) $\sum_{n=1}^{\infty} u_n$ B) $\sum_{n=1}^{\infty} (\sin n) u_n$ C) $\sum_{n=1}^{\infty} \frac{u_n}{n}$ D) $\sum_{n=1}^{\infty} n u_n$